Human-Computer Interaction

Agents, Delegation & Embodiment

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Today's Agenda

- Topic overview: *Agents, Delegation & Embodiment*
- The frame: *enduring fundamental → how AI changes it*
- Provocation: *Automate the sequence, or keep initiative returnable?*
- Discussion

This term's frame

Enduring fundamental → *How AI is changing it* → *Depth (a claim to argue with)*

Fundamentals are not a relic to move past — they are the **instrument** you use to read the AI moment.

- **Today's fundamental:** *delegation* — a **transfer of authority** to act on a principal's behalf, over a *sequence* of actions
- **Depth: Deep** — software agents and embodied robots are **one family**: reliance over action sequences

Topic overview: *Agents, Delegation & Embodiment*

Delegation as transfer of authority

The enduring question isn't "is it an agent?" — it's *what did you hand over, and can you get it back?*

- Delegation = authority to act **on your behalf**, over a **sequence** of actions
- Software or embodied — **one family**, one problem: *reliance over sequences*
- The hard sub-questions: **agency · autonomy · accountability · *when* to delegate**

Software agents & agency

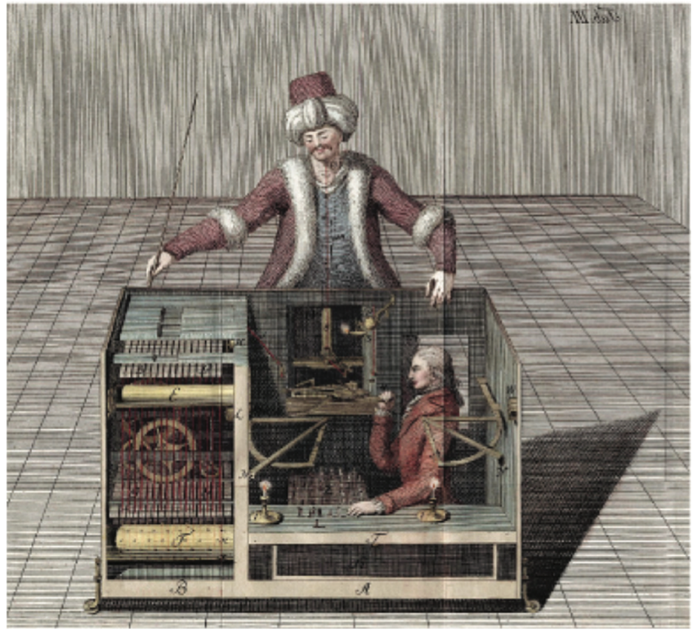
Software agent: a program that acts for a user or another program in a relationship of *agency*.

Agency: an agreement to act on one's behalf.

Agency implies intelligence, autonomy, and decision-making – and, with them, accountability.

A long lineage of "artificial others"

- **1495 / 18th c.:** Leonardo's mechanical knight; the Turk, a human hidden inside
- **1950s–90s:** programmable robot arms → autonomous software agents
- **1990s–2000s:** embodied conversational agents, intelligence *located* in a body
- **2020s:** agentic AI, LLM agents that plan, use tools, remember





Anchor: agency is *relational*[1]

Lucy Suchman — *Human-Machine Reconfigurations* (2007)

- Agency is a **relational/distributed achievement**, not a property inside the machine
- "Autonomy" is sustained by **hidden human labor** — the antidote to "the agent acts on its own"
- Every "the agent did it alone" claim → *ask what human work was made invisible*

[1] Suchman, L. (2007). *Human-Machine Reconfigurations: Plans and Situated Actions*, 2nd ed. Cambridge Univ. Press. *(Recurs Wk 13.)*

Initiative, made *returnable*[2]

Wendy Mackay – co-adaptation & human-computer partnerships

- Initiative is **shared**, made **visible**, and can be **handed back**
- Humans and systems **mutually adapt** over time
- The design test: after the agent acts, can the human *take the wheel again* – cleanly?

[2] Mackay, W.E. (2024). Creating Human-Computer Partnerships. In *Computer-Human Interaction Research and Applications (CHIRA 2023)*, CCIS vol. 1996, pp. 3–17. Springer. doi:10.1007/978-3-031-49425-3_1

Reliance over *sequences*[3]

Gajos & Mamykina – *Do People Engage Cognitively with AI?* (IUI 2022)

- Do people think *with* delegated assistance – or *defer* to it?
- AI assistance can suppress **incidental learning** – you get the answer, not the understanding
- Reliance is measured over **decisions**, not a single call

[3] Gajos, K.Z. & Mamykina, L. (2022). Do People Engage Cognitively with AI? Impact of AI Assistance on Incidental Learning. *IUI 2022*.

Agents as a research instrument[4]

Park et al. – *Generative Agents: Interactive Simulacra of Human Behavior* (UIST 2023)

- Software agents as a *research instrument* for delegation & multi-agent behavior
- *(You also read this Wk 5 – CSCW; re-read here through the delegation/embodiment lens)*
- Caution: simulacra of aggregate behavior, **not** evidence about

[4] Park, J.S., O'Brien, J., Cai, C.J., Morris, M.R., Liang, P. & Bernstein, M.S. (2023). Generative Agents. *UIST 2023*.

Neither assistant nor butler[kn]

The 1987 video you saw in Week 1 — coded with three frameworks, asking *why doesn't this exist yet?*

- **Three of four barrier families are HCI problems** — privacy, social/situational, trust — only the fourth is technological
- **Authority is asymmetric:** Phil can interrupt Mike, unmute himself, and speak for him. Mike mutes Phil by *tapping his face*
- **We lack a word for it:** not assistant, not butler — Mike never has

[kn] Newendorp, A.K., Sanaei, M., Perron, A.J., Sabouni, H., Javadpour, N., Sells, M.J., Nelson, K.T., Dorneich, M.C. & Gilbert, S.B. (2024). Apple's Knowledge Navigator: Why Doesn't that Conversational Agent Exist Yet? *CHI '24*, Article 776, 14 pp. doi:10.1145/3613904.3642739

Why does embodiment matter?[5]

"We need to locate intelligence… The best example of located intelligence, of course, is the body." – Cassell, 2001

A body gives us:

- A **locus of attention** – a target for attention & behavior
- Cues about the agent's **status** (working, speaking, waiting)
- A coherent **character** signaling role (butler, assistant,

[5] Cassell, J. (2001). Embodied conversational agents: representation and intelligence in user interfaces.

Embodiments → frames of mind[6]

Physical frame

Co-situated; proxemic norms; seen as an **independent agent**; engagement is voluntary.

Virtual frame

User enters a crafted plot; conventions are learned; interaction is richly scripted.

Also: blended, mediated, immersive frames. Embodiment is the **high-stakes axis** — physical, irreversible action.

[6] Mutlu, B. (2021). The virtual and the physical: two frames of mind. *iScience*, 24(2).

Physical vs. virtual: the tradeoffs

Mechanism	Physical	Virtual
Situativity	Co-situated in *your* environment	You are brought into *its* environment
Agency	Independent agent, own goals	Engagement at the user's discretion
Proxemics	Dynamic, co-managed to human norms	Constrained, learned conventions
Believability	Real-world, self-relevant	Safe space to experience emotion

Method callback: Wizard-of-Oz[7]

(From M3 — carried into Friday/Studio work.)

- **The** method for prototyping agent behavior *before the model exists*
- A human "wizard" drives the agent; users experience the interaction, not the plumbing
- Lets you study delegation & initiative-return **before** committing to autonomy

[7] Kelley, J.F. (1984). An Iterative Design Methodology for User-Friendly Natural Language Office Information Applications. *ACM Transactions on Office Information Systems*, 2(1), 26–41. doi:10.1145/357417.357420 *(the WoZ anchor from M3.)*

But what about *AI*?

LLM agents change the *cost* of delegation

Tool-calling, computer-use, multi-agent — delegation becomes **cheap, recursive, opaque**.

- **Cheap:** delegate whole sequences with a prompt
- **Recursive:** agents spawn agents; authority forwards without oversight
- **Opaque:** the sequence of actions is hard to see, harder to *return* from

👉 Software agents and HRI/teleoperation are now **one problem** —

Guardrails coda

- **Suchman:** "autonomy" conceals the human-in-the-loop labor that sustains it
- **Accountability** must hold over a *sequence*, not a single decision
- **Embodiment** raises the stakes: physical, *irreversible* action demands a returnable path

Your provocation *(post one slide)*

Pick one:

1. **Provocation** — the tradeoff: *efficiency-through-delegation* vs. *preserved human agency & accountability* ("automate the sequence" vs. "keep initiative visible & returnable"). Ground it in the readings.
2. **Critical artifact** — a real AI agent (software or embodied). Annotate where authority is transferred, where initiative is (or isn't) *returnable*, and where "autonomy" hides human-in-the-loop labor.

Discussion

Today's roles

Reporter	Skeptic	Connector
Seat 1	**Seat 2**	**Seat 3**

Same in every pod. **Seats 4, 5 and 6** have no role today — you are still in the argument.

Discussion Format

In your pod, on your pod monitor — ~30 minutes.

- **Everyone shows their slide** — about 2 minutes each, cast from your own laptop
- **Skeptic** puts a counterargument on the table · **Connector** ties it to an earlier week
- **Reporter** writes the pod's three lines — agreed, split, unresolved — and delivers them to the room

What's Next?

- **Friday – Methods: Statistics I (Descriptive + Effect Sizes & CIs)**
  - **Read:** the textbook **Chapter 4** (descriptive statistics) · Cumming (2014), *The New Statistics*
  - Post your team's **3-line project status** (done / blocker / next)
- *Methods handoff:* claim "users delegate appropriately" → measure appropriate / over-reliance across **agent-correctness** conditions, with accountability & initiative-return as manipulations *(seeds your evaluation design)*.